Prithvi Chakravarthy

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Education

DePaul University — Chicago, IL
Master of Science in Computer Science — *Sept. 2025 – Present*

- Coursework: Distributed Systems, Applied Algorithms and Structures, DBMS, Real-Time System Architecture

BITS Pilani, WILP — India
Postgraduate Certificate in AI/ML — *Oct. 2023 – Oct. 2024*

- Coursework: Deep Learning, Neural Networks, Text Mining, Feature Engineering, Unsupervised Learning

VIT Bhopal — India
Bachelor of Technology in Computer Science — *Jun. 2018 – Jun. 2022*

- Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Computer Architecture, Soft Computing, Computer Networks, Theory of Computation & Compiler Design

Experience

Designare Solutions Pvt Ltd — Mumbai, India
Full Stack Developer — *March 2022 – August 2025*

- Developed and deployed full stack applications using React, Node.js, Python and PostgreSQL, improving scalability and performance.
- Designed distributed APIs and optimised data pipelines, reducing query time by 25%.
- Managed production deployments on Google Cloud, maintaining 99.9% uptime.
- Implemented secure user authentication and role-based access control (RBAC).
- Collaborated with stakeholders to define requirements and deliver on time.

Projects

Space Invaders Game Engine | *C#, Azul Engine, IrrKlang*

- Engineered a high-performance Space Invaders clone using **C#** and the **Azul engine**, strictly adhering to **SOLID principles** and implementing **14+ GoF design patterns** to ensure a modular and decoupled architecture.
- Optimized memory management by architecting a universal **Object Pool** system (Template pattern), eliminating runtime heap allocation spikes and leveraging **Proxy/Flyweight patterns** to reduce GPU memory overhead for sprite rendering.
- Developed a decoupled collision detection system using **Visitor and Observer patterns** for double-dispatch resolution, and managed interval-based game events via a **Command pattern** paired with a **Priority Queue**.
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IQGeo (Telecom/GIS Platform) | *Python, PyTorch, Jupyter*

- Developed backend APIs and React modules to visualize large geospatial datasets, translating client requirements into technical specifications.
- Architected scalable Python (Pyramid) APIs to handle complex spatial queries, improving data retrieval efficiency for telecom clients.

Stable Diffusion Image Generation Model | *Python (Pyramid), JavaScript*

- Developed a latent diffusion model using PyTorch to generate high-fidelity images from text prompts, focusing on efficient tensor operations and model architecture.
- Implemented and optimized U-Net and attention mechanisms to improve inference speed and generation quality.
- Implemented a scene-driven game loop using the **State pattern** to manage transitions between Select, Play, and Game Over states, ensuring seamless real-time interactivity.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, Kotlin, C#
Frameworks/Libraries: React, Node.js, Pyramid, MaterialUI, TailwindCSS, PyTorch, TensorFlow
Databases: PostgreSQL, MySQL, Oracle
Tools: Google Cloud, GitHub, Docker, Jenkins, CI/CD, REST APIs, Jest